Identifying the Most Influential Predictors of a US Presidential Election Outcome

(Callixta Cahyaningrum, Elton Chow, Sydney Wood, Kent Arifin)

I. Abstract

With the U.S. Presidential Election occurring within the past semester, we aim to analyze how various predictors such as population size, money raised, crime rate, education and economic stability may contribute to the election outcome to answer our research question of *What factors may account for the winner of a U.S. Presidential Election?* These variables were chosen based on their potential to reflect socio-economic, financial, and political conditions that influenced voter behaviour and were sourced from websites such as opensecrets.org and worldpopulationreview.com. Using tools such as simple linear regression, multiple linear regression, logistical regression, etc. in RStudio, we identified that money raised by either political party generally plays the biggest role in determining the presidential election.

II. Introduction

Every four years, the world sits back and waits for the results of the United States presidential election. Being one of the most influential countries in the whole world, it can be said the result will ultimately decide how the world lives for the following four-year period. CSIS (2024) argues that when an American goes to the polls to vote, their choices have consequences for the world. This ultimate decision often depends on certain influential variables that impact the individual's life, from the state they live in, their beliefs and other demographics Gelman & King (2009). So forth, this paper aims to explore and analyze distinct predictor variables that play a role in answering our research question of *What factors may account for the winner of a U.S. Presidential*

Election? Although some studies have also attempted and concluded on this topic, most of these studies like Lichtman (2024) and Farnschläder (2024) are either inaccurate or use a different form of statistical analysis. Not to mention, most of the available studies like Sinha et al. (2012) are outdated and do not reflect the evolving nature of the world, in addition to this year's unprecedented U.S. Presidential Election. Of those related to this year's election, most of them were not credible as they were made by broadcasting companies which have proven to be somewhat biased. To address this deficit, this paper approaches this significant knowledge gap by attempting to find and create our best model for predicting the most influential variable of the 2024 U.S. Presidential Election using regression analysis. From this, this paper argues that for both Democrats and Republicans, the amount of money raised is the key to 270 electoral votes. Furthermore, it contributes to the study of politics and opens up many interesting avenues for exploration: a) how different the set of predictor variables may change and determine the outcome, b) what other predictor variables may determine the result and c) from our limitations such as bias and external validity, how other potential statistical models may predict the result differently. The remainder of the paper is as follows: First, we will expand on previously mentioned studies in our state of knowledge. Secondly, we will define and explain our chosen variables, in addition to the sources, of our model. Then we will dive deep into our analysis and discuss the results of the models used. Lastly, we will close by stating our conclusion, limitations, and contributions and offer advice for future research before wrapping it up with a reflection on the process of completing this investigation.

III. Review of Previous Literature

As CSIS (2024) mentions, the audience of the US presidential election is global and their choices have consequences for the world. This is due to the potential policies and beliefs their president-elect may impose and have, in addition to the US stance on all global issues. From China's assertiveness, the Russia-Ukraine war, Israel-Palestine war to the overall international order. Moreover, the political party of the elected president may have different stances on the mentioned affairs. For instance, a democrat president tends to support progressive reforms through redistributive social and economic policies whereas a republican president tends to restore traditional institutions and limit change (Cooper & Vahedyousefi, 2024). Given this, there have been multiple studies attempting to predict the results of a presidential election. However as mentioned most of these studies are either misguided, outdated or use a different form of statistical analysis. or misguided. For example, Sinha et al (2012) used multiple regression to predict the 2012 U.S. presidential election of Obama vs Romney; Lichtman (2024) used an index system to predict Kamala Harris to win the 2024 U.S. presidential election; and Farnschläder (2024) used machine learning to predict that a very close election with a slight edge with Kamala Harris winning the 2024 U.S. presidential election.

IV. Specification of the Model

In the following model, 5 key variables were chosen to predict the 2024 U.S. presidential election: Money raised (by each party), unemployment rate, crime rate, education rate, and population, which were compared to the number of votes for each party. Each variable was carefully chosen after research (Gelman & King, 2009). Although we chose to use these five predictor values, there are countless other potential variables that may influence and account for the final election outcome, such as income

disparity, population density, and occupation to name a few. Nonetheless, we felt that the chosen five were the most important as they represent a fairly comprehensive range of American identity demographics. We do, however, expect a difference in how impactful each variable will be in influencing the final outcome. We've listed our expectations for all 5 variables, followed by our reasoning/assumptions for each prediction.

- Money raised : for every dollar raised for the given party, we expect votes to increase for the same party. This means that we expect a non-negative mathematical operator in the correlation matrix cells between money raised and votes for *both* democrats and republicans.
- Unemployment rate (representing economic standing) : for every percent increase in unemployment, we expect votes for democrats to increase and votes for republicans to decrease, due to political trends around each party's stance on welfare. This means we expect a non-negative mathematical operator in the democratic correlation matrix cells between unemployment rate and money raised, and we expect a negative mathematical operator in the republican correlation matrix for the same cells.
- Crime rate : for every percent increase in crime rate, we expect votes for democrats to increase and votes for republicans to decrease, due to democrats' historically greater emphasis on systematic social reform. This means we expect a non-negative mathematical operator in the democratic correlation matrix between crime rate and votes, and we expect a negative mathematical operator in the republican correlation matrix for the same cells.
- Education rate : for every percent increase in the rate of individuals obtaining a bachelor's degree or higher, we expect votes for democrats to increase, whereas

votes for republicans to decrease, as higher education levels are often associated with liberal social and political views. This means we expect a non-negative mathematical operator in the democratic correlation matrix between education rate and votes, and we expect a negative mathematical operator in the republican correlation matrix for the same cells.

- Population : for every additional individual, we expect votes to increase for both parties. We expect this solely because more people generally means a greater proportion of those people being capable of voting (meaning of voting age and bearing the inclination to fill out a ballot). This means that we expect a non-negative mathematical operator in the correlation matrix cells between money raised and votes for *both* democrats and republicans.

V. Data Description

In our CSV file are various variables that require further explanation:

- money_raised_dem is the quantity of money raised by democrats (in $ USD)
- votes_dem is the quantity of votes for democrats (# of votes counted)
- money_raised_rep is the quantity of money raised by republicans (in $ USD)
- votes_rep is the quantity of votes for republicans (# of votes counted)
- population is the quantity of people meeting the given criteria (# of people)
- education is the quantity of people per state that received a Bachelor's degree or higher (# of people)
- education rate is the quantity of individuals per state meeting the education criteria divided by the total population of the state (% of the total population in the state)
- economy is the quantity of unemployed individuals per state (# of people)

- economy_rate is the quantity of unemployed individuals per state divided by the total population of the state (note this title was changed in R as unemployment_rate) (% of the total population in the state)
- crime is the quantity of crimes that were reported in a single year (# of crimes committed)
- crime_rate is the quantity of crimes divided by the total population of the state (% of the total population in the state) - we made the assumption that each crime could be attributed to one single person, thus making the number of reported crimes equal to that number of criminals.
- state is each of the 50 Continental US States
- win_dem is the quantity of votes that went to democrats, which we found for each state. Note that this title was created in R for logistic regression, and for linear regression, our label is votes_dem. (# democratic votes counted).
- win_rep is the quantity of votes that went to republicans, which we found for each state. Note that this title was created in R for logistic regression, and for linear regression, our label is votes_rep. (# republican votes counted)
- money_raised_share_dem is the quantity of money raised by the democratic party divided by the total quantity of money raised by both parties. Note that this title was created in R for logistic regression only because the model does not depend on raw money count. (% money raised by democrats)
- money_raised_share_rep is the quantity of money raised by the republican party divided by the total quantity of money raised by both parties. Note that this title was created in R for logistic regression only because the model does not depend on raw money count. (% money raised by republicans)

The data that we used for our variables were gathered from a variety of credible sources: worldpopulationreview.com, opensecrets.org, bls.gov, and uselectionatlas.org. These websites provide extensive databases covering, for each state, detailed demographic data, economic data, political contributions and election results, which are essential for our analysis of the factors influencing U.S. Presidential election outcomes. However, it is imperative to mention certain limits within our aggregated data set such as potential biases in reporting, inconsistencies in data collection methods across different sources, and most importantly, the time-sensitivity of the collected data. Awareness of any time dependency is especially crucial because data such as crime and education rates are ever-changing. Additionally, while our sources are valuable and provide practical data for our purposes, because of the scope of this project, we only considered 5 variables and pulled data for each variable from one source; this means that our model does not capture all nuances of regional variations or the complete spectrum of socio-economic factors that can influence electoral outcomes. These limitations may affect the generalizability of our findings and should be taken into account when interpreting the results.

VI. Results

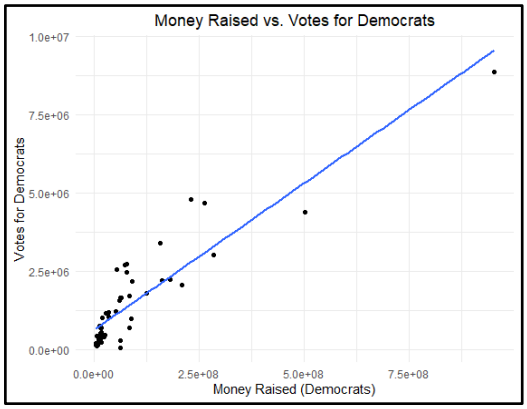
Checking the Relationships and Assumptions of the Model

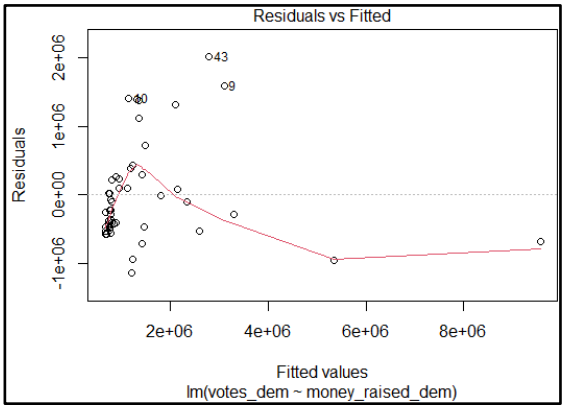
To start, we shall determine the relationship and check three major assumptions of the linear regression model: a) Homoscedasticity - we can check this using a residual plot and confirm if the residuals are spread equally along the x-axis; b) Linearity-we can check this using a scatter plot and confirm if the data points are linear; c) Normality-we can check this using a Q-Q plot and confirm if the residuals are linear; and d)

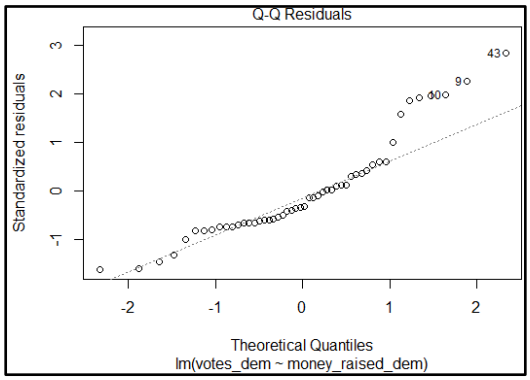
Independence. Given these, the following are 3 plots for each independent variable against the dependent variable, for each party:

DEMOCRATIC

Money Raised vs Votes for Democrats

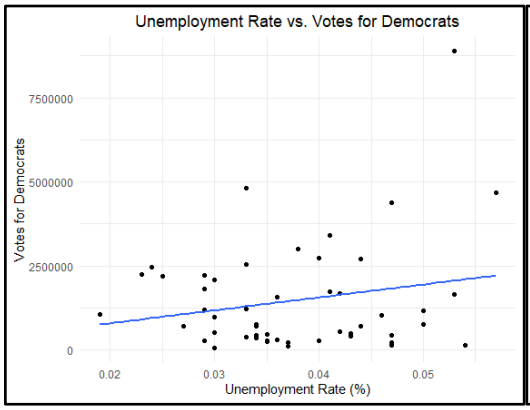


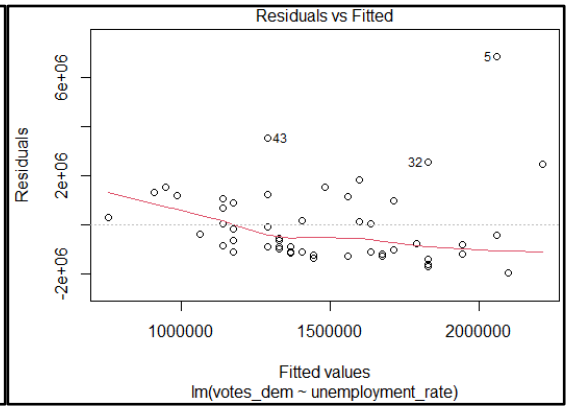


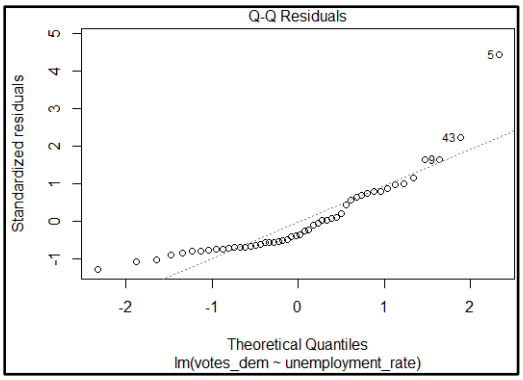


- The scatterplot (top left) shows a strong positive linear relationship
- Residuals vs Fitted (top right) shows a heavy level of heteroskedasticity
- Q-Q Residuals (bottom left) shows that the residuals are not normally distributed

Unemployment Rate vs Votes for Democrats

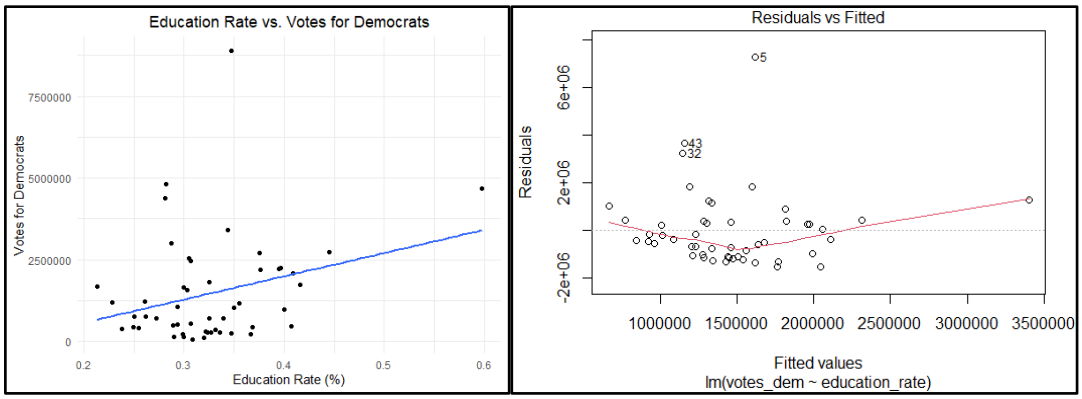


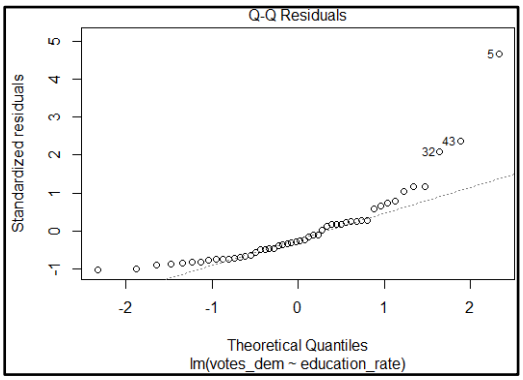




- Scatterplot (top left) shows a weak positive linear relationship
- Residuals vs Fitted (top right) shows profound heteroskedasticity
- Q-Q Residuals (bottom left) shows that the residuals are not normally distributed

Education Rate vs Votes for Democrats





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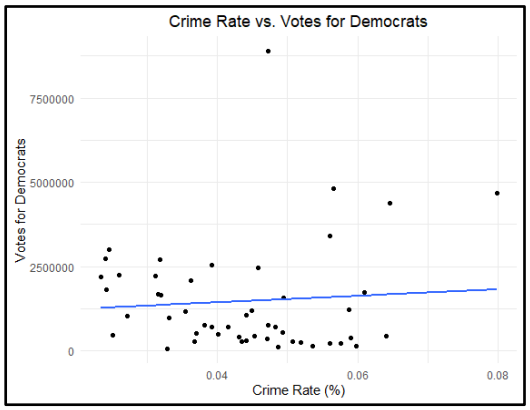
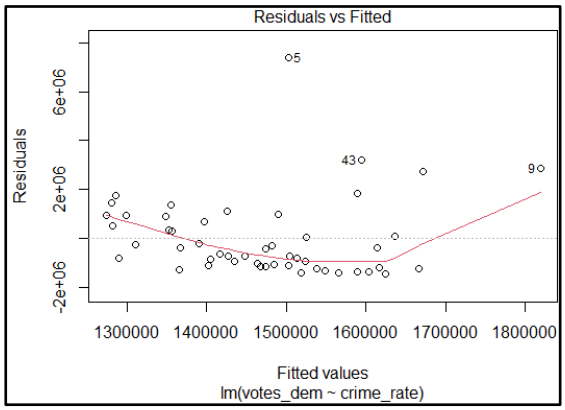
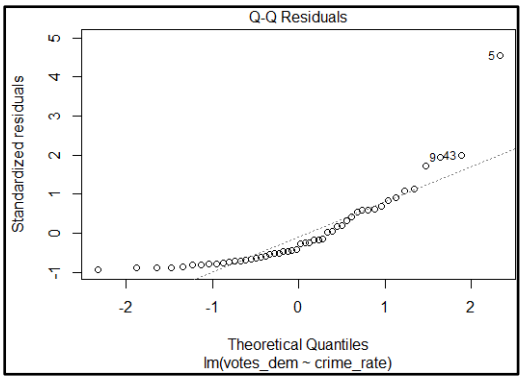
Crime Rate vs Votes for Democrats

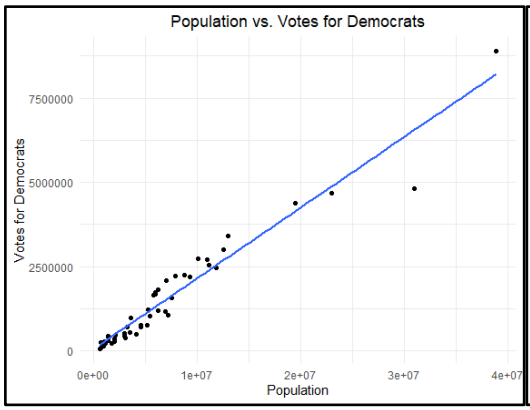


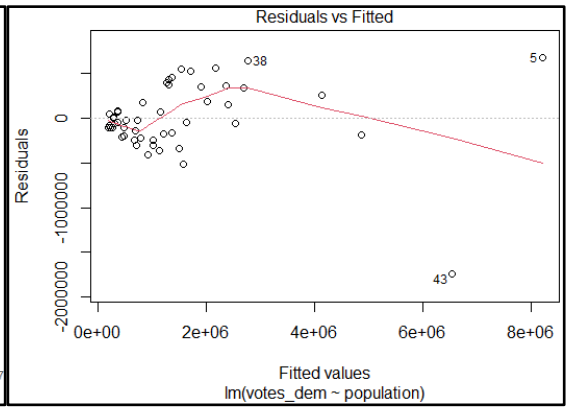


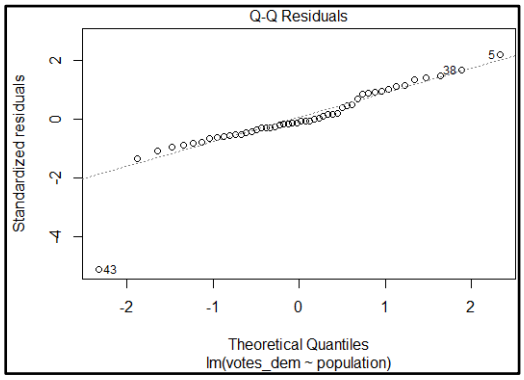


- Scatterplot (top left) shows a very weak positive linear relationship
- Residuals vs Fitted (top right) shows profound heteroskedasticity
- Q-Q Residuals (bottom left) shows that the residuals are not normally distributed

Population vs Votes for Democrats







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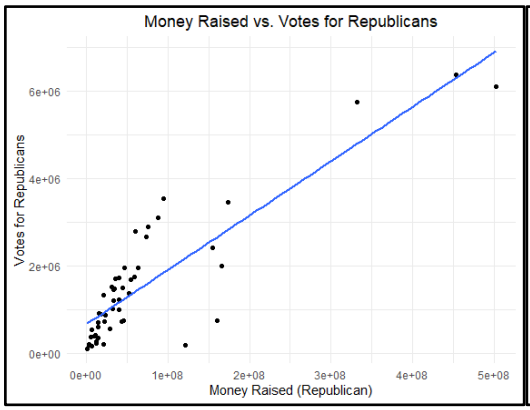
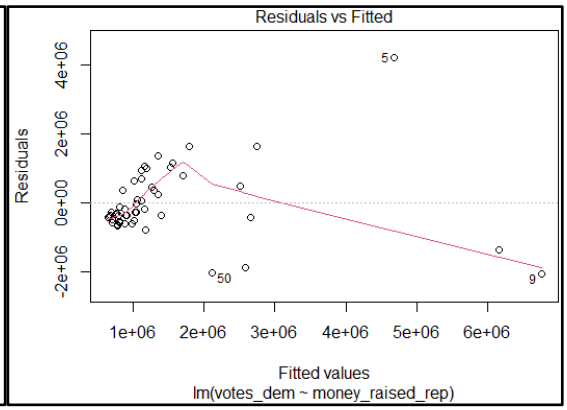
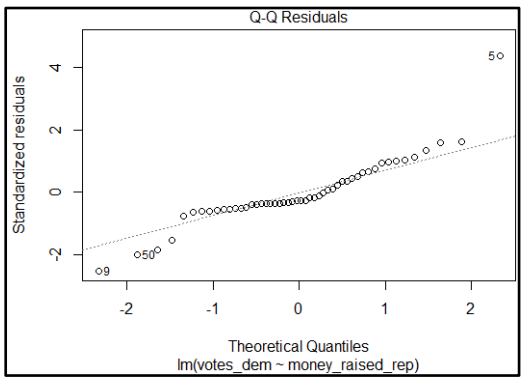
REPUBLICAN

Money Raised vs Votes for Republicans

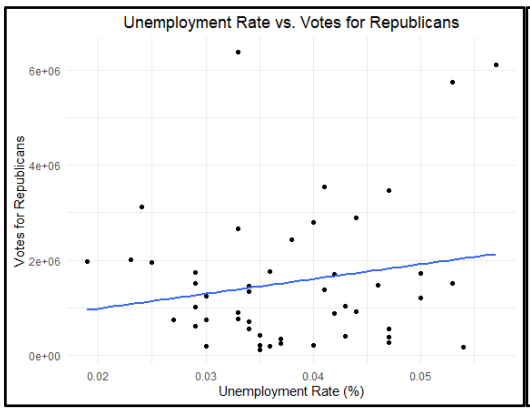


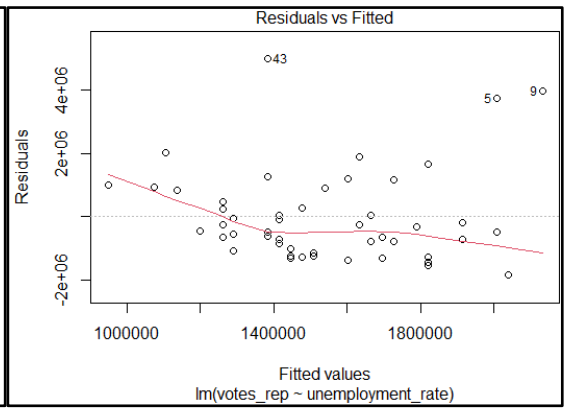


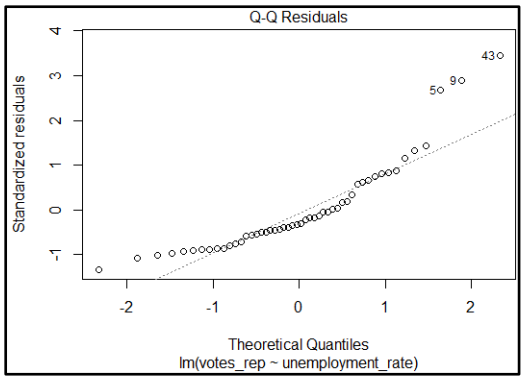


- Scatterplot (top left) shows a strong positive linear relationship
- Residuals vs Fitted (top right) shows profound heteroskedasticity
- Q-Q Residuals (bottom left) shows that the residuals are not normally distributed

Unemployment Rate vs Votes for Republicans







- Scatterplot (top left) shows a weak positive linear relationship
- Residuals vs Fitted (top right) shows profound heteroskedasticity
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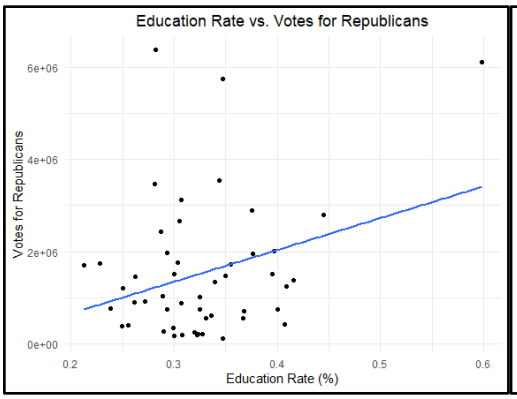
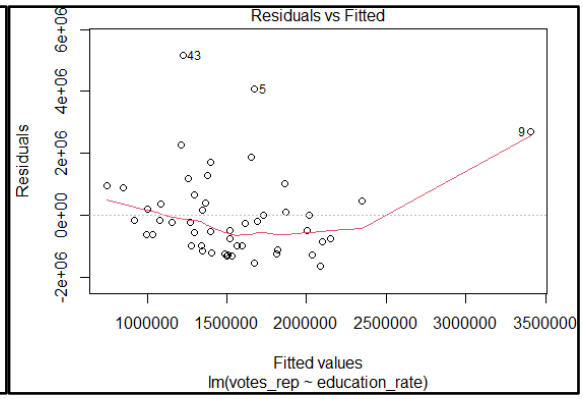
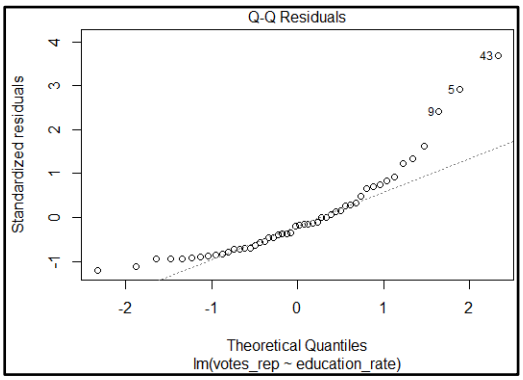
Education Rate vs Votes for Republicans

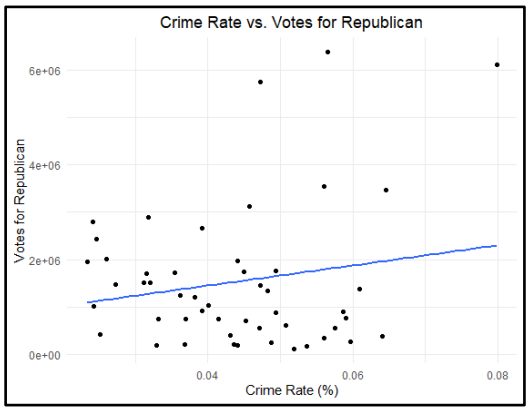


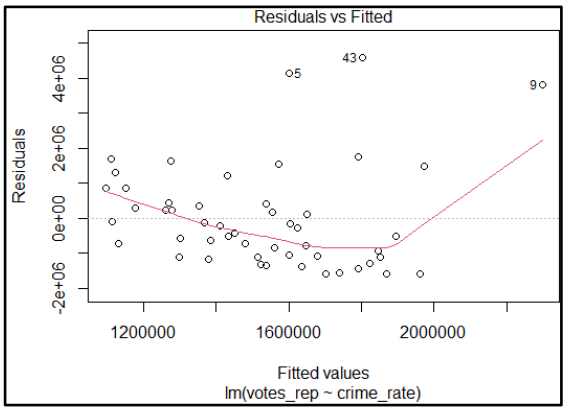


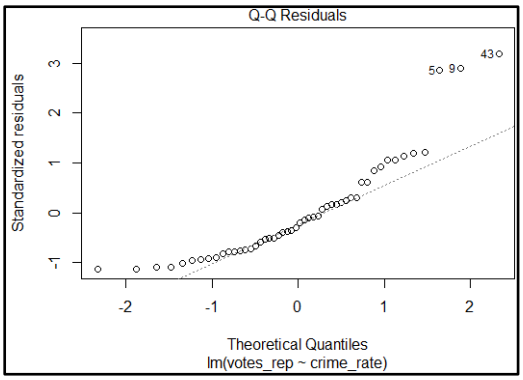


- Scatterplot (top left) shows a fair positive linear relationship
- Residuals vs Fitted (top right) shows profound heteroskedasticity
- Q-Q Residuals (bottom left) shows that the residuals are not normally distributed

Crime Rate vs Votes for Republicans







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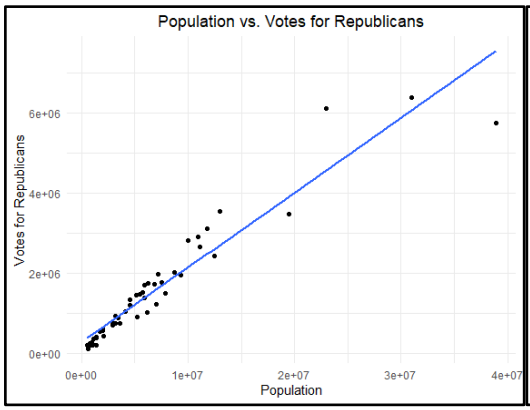
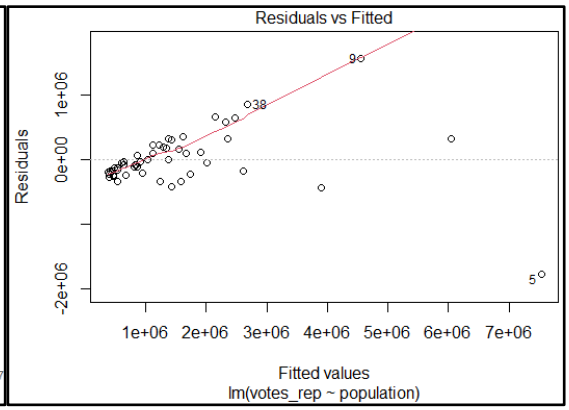
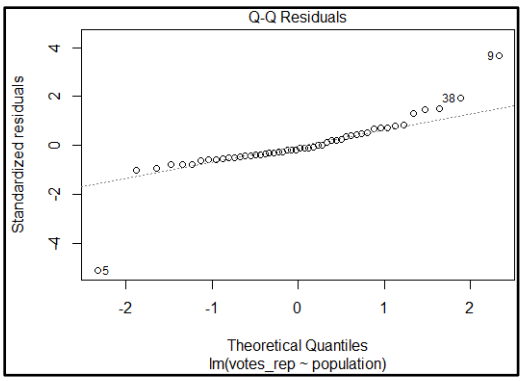
Population vs Votes for Republicans







- Scatterplot (top left) shows a strong positive linear relationship
- Residuals vs Fitted (top right) shows a nonlinear heteroskedasticity
- Q-Q Residuals (bottom left) shows residuals are not normally distributed

Linear Regression

Note that backward stepwise regression was not done since we focus on the context of econometric modelling, which would be more focused on theoretical reasoning and model interpretability rather than purely statistical ‘optimization’.

Democrats (figures labelled below)

```
> regl_dem<-lm(votes_dem~money_raised_dem + unemployment_rate + crime_rate + education_rate + population, data=data)
> summary(regl_dem)

Call:
lm(formula = votes_dem ~ money_raised_dem + unemployment_rate +
    crime_rate + education_rate + population, data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-742442 -143260  -36290  157195  833912

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)       -1.425e+05  2.874e+05  -0.496   0.6224
money_raised_dem   2.643e-03  5.060e-04   5.222 4.61e-06 ***
unemployment_rate  5.156e+06  5.078e+06   1.015   0.3155
crime_rate        -8.920e+06  3.502e+06  -2.547   0.0144 *
education_rate     1.498e+06  6.504e+05   2.303   0.0261 *
population         1.613e-01  1.058e-02  15.244  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 288200 on 44 degrees of freedom
Multiple R-squared:  0.9717,    Adjusted R-squared:  0.9685
F-statistic: 301.9 on 5 and 44 DF,  p-value: < 2.2e-16
```

SLR Summary

```
> cor_dem<-cor(data[c("votes_dem", "money_raised_dem", "unemployment_rate", "crime_rate", "education_rate", "population")])
> cor_dem
                  votes_dem money_raised_dem unemployment_rate   crime_rate education_rate population
votes_dem         1.00000000       0.89913655         0.2075147  0.075042452    0.285708512  0.9712904
money_raised_dem  0.89913655       1.00000000         0.2175669  0.087200987    0.190514228  0.8484053
unemployment_rate 0.20751473       0.21756687         1.0000000  0.324986020    0.076303095  0.1905383
crime_rate        0.07504245       0.08720099         0.3249860  1.000000000   -0.009470499  0.1521781
education_rate    0.28570851       0.19051423         0.0763031 -0.009470499    1.000000000  0.2333321
population        0.97129039       0.84840525         0.1905383  0.152178090    0.233332147  1.0000000
```

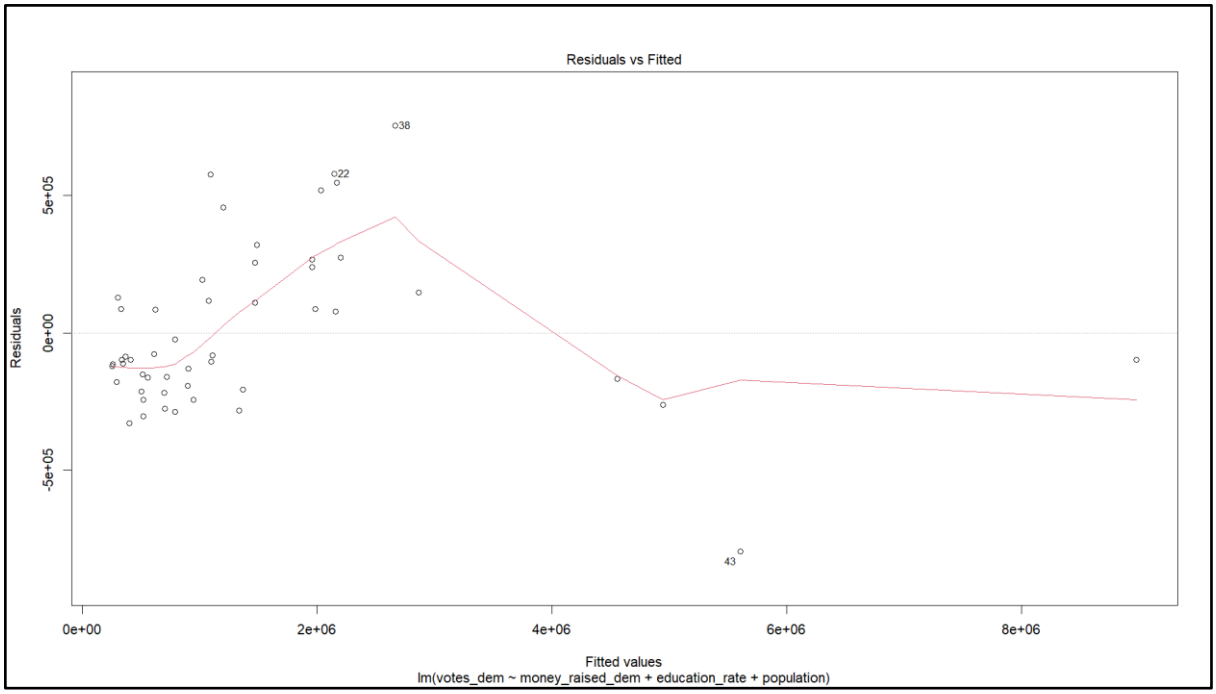
Correlation Matrix

```
> vif(regl_dem)
 money_raised_dem unemployment_rate        crime_rate    education_rate        population
         3.665066           1.170031          1.149533          1.063258          3.733273
```

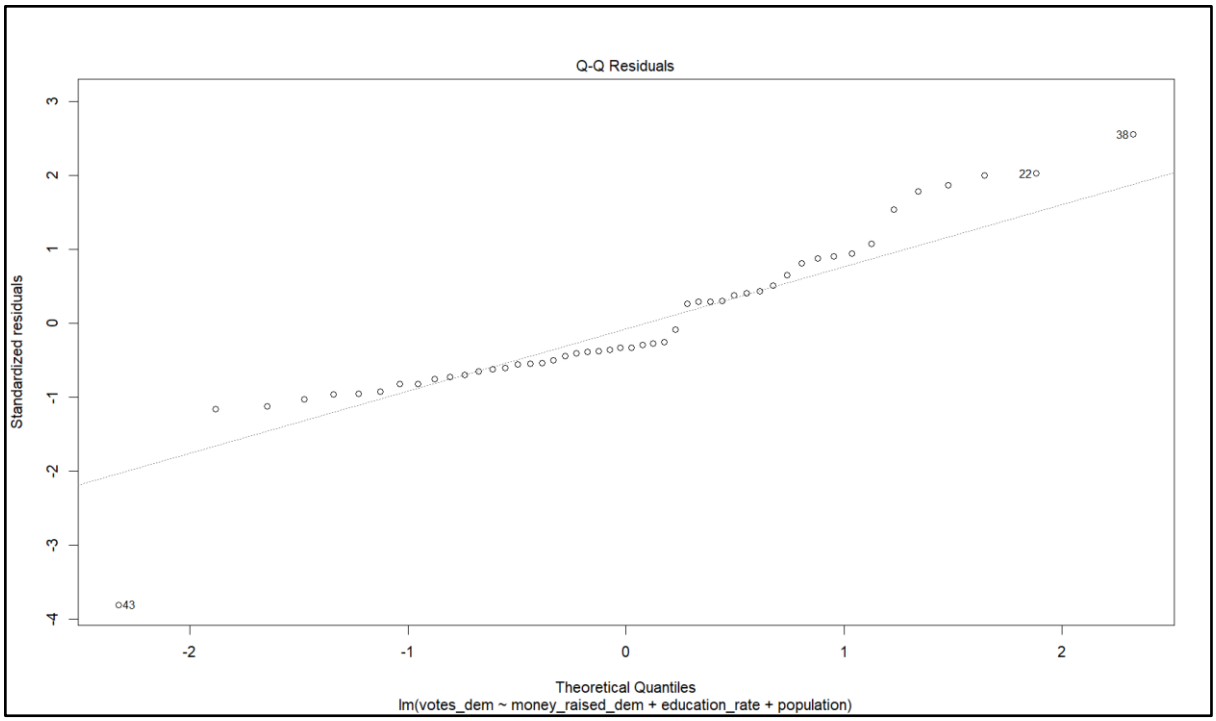
Variance Inflation Factor

```
> bptest(regl_dem)

        studentized Breusch-Pagan test

data:  regl_dem
BP = 22.535, df = 3, p-value = 5.049e-05
```

Test for Heteroskedasticity



Residual vs Fitted Plot



QQ Plot

Republican (figures labelled below)

```
> regl_rep<-lm(votes_rep~money_raised_rep + unemployment_rate + crime_rate + education_rate + population, data=data)
> summary(regl_rep)

Call:
lm(formula = votes_rep ~ money_raised_rep + unemployment_rate +
    crime_rate + education_rate + population, data = data)

Residuals:
     Min       1Q   Median       3Q      Max
-1436374  -237411   -90372   232172   940943

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)       -1.845e+05  4.538e+05  -0.407   0.6863
money_raised_rep   2.466e-03  1.243e-03   1.984   0.0535 .
unemployment_rate -3.651e+05  7.459e+06  -0.049   0.9612
crime_rate         1.732e+06  5.441e+06   0.318   0.7518
education_rate     1.344e+06  1.021e+06   1.316   0.1950
population         1.548e-01  1.591e-02   9.728 1.55e-12 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 424000 on 44 degrees of freedom
Multiple R-squared:  0.9254,    Adjusted R-squared:  0.9169
F-statistic: 109.2 on 5 and 44 DF,  p-value: < 2.2e-16
```

Simple Linear Regression

```
> cor_rep<-cor(data[c("votes_rep", "money_raised_rep", "unemployment_rate", "crime_rate", "education_rate", "population")])
> cor_rep
                  votes_rep money_raised_rep unemployment_rate   crime_rate education_rate population
votes_rep         1.0000000        0.8692979         0.1852965  0.182317797    0.306136001  0.9536516
money_raised_rep  0.8692979        1.0000000         0.1581312  0.279637795    0.359770373  0.8482289
unemployment_rate 0.1852965        0.1581312         1.0000000  0.324986020    0.076303095  0.1905383
crime_rate        0.1823178        0.2796378         0.3249860  1.000000000   -0.009470499  0.1521781
education_rate    0.3061360        0.3597704         0.0763031 -0.009470499    1.000000000  0.2333321
population        0.9536516        0.8482289         0.1905383  0.152178090    0.233332147  1.0000000
```

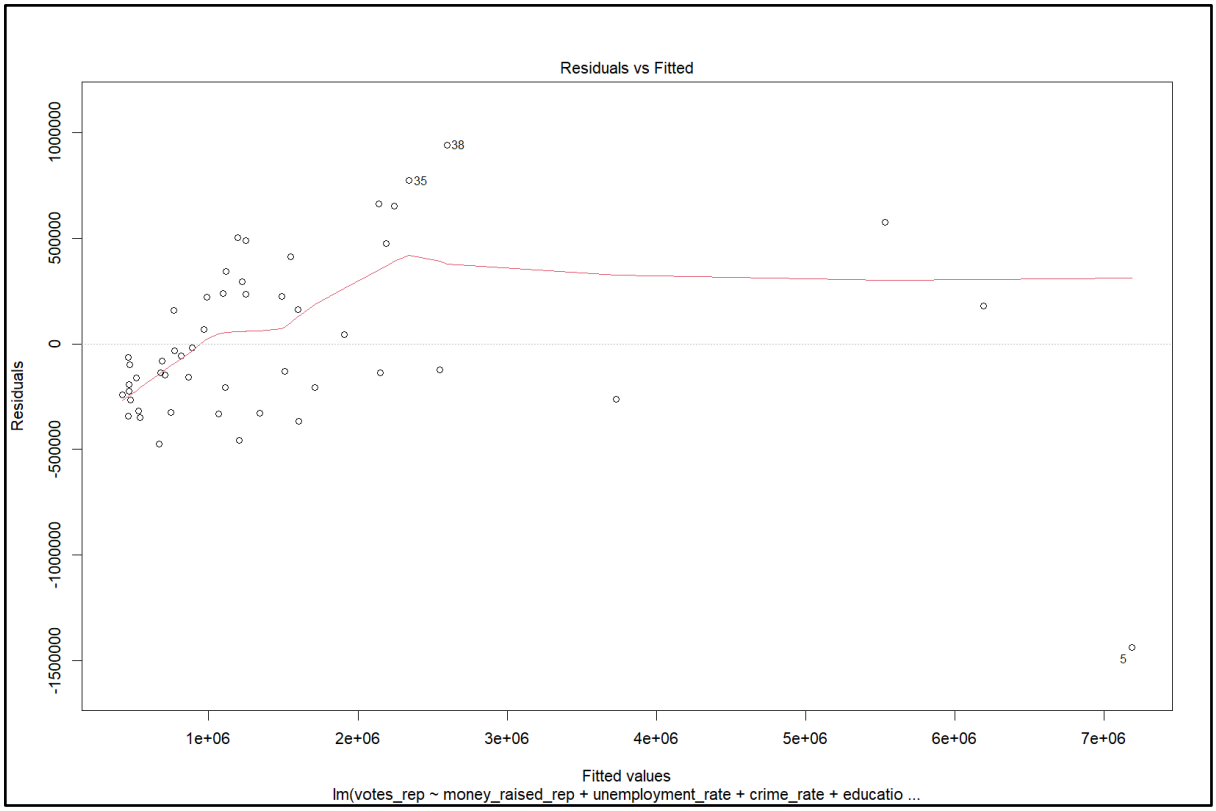
Correlation Matrix

```
> vif(regl_rep)
 money_raised_rep unemployment_rate         crime_rate     education_rate         population
         4.477270           1.166333           1.281779           1.210965           3.902987
```

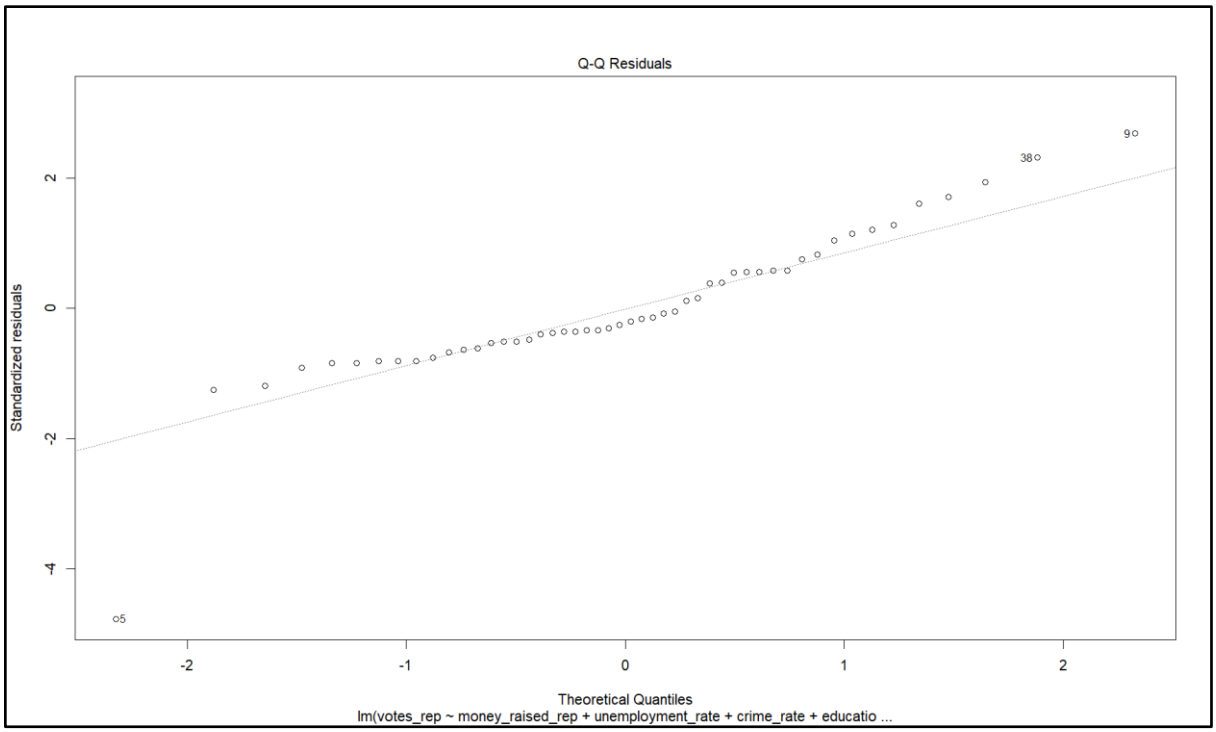
Variance Inflation Factor

```
> bptest(regl_rep)

        studentized Breusch-Pagan test

data:  regl_rep
BP = 25.341, df = 3, p-value = 1.31e-05
```

Test for Heteroskedasticity



Residual Plot for the Linear Regression Model



QQ Plot for the Linear Regression Model

Logistic Regression

Note that logistic regression was attempted however the end result was not as desired. Thus we believe that our linear regression model is the best model. However, we can analyze the power of money in predicting whether the democratic or the republican electoral candidate will win the election using this model.

DEMOCRAT

```
> #overall model (which has terrible p value)
> logit_dem<- glm((win_dem) ~ money_raised_share_dem + unemployment_rate + crime_rate + education_rate + population
+                 data = data, family = "binomial")
Warning messages:
1: glm.fit: algorithm did not converge
2: glm.fit: fitted probabilities numerically 0 or 1 occurred
> summary(logit_dem)

Call:
glm(formula = (win_dem) ~ money_raised_share_dem + unemployment_rate +
    crime_rate + education_rate + population, family = "binomial",
    data = data)

Coefficients:
                        Estimate Std. Error z value Pr(>|z|)
(Intercept)            -8.049e+02  2.355e+05  -0.003    0.997
money_raised_share_dem  1.805e+03  5.208e+05   0.003    0.997
unemployment_rate      -7.479e+03  2.401e+06  -0.003    0.998
crime_rate             -1.682e+03  6.560e+05  -0.003    0.998
education_rate          3.396e+02  1.683e+05   0.002    0.998
population             -3.978e-06  1.397e-03  -0.003    0.998

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 6.6406e+01  on 49  degrees of freedom
Residual deviance: 2.2811e-08  on 44  degrees of freedom
AIC: 12

Number of Fisher Scoring iterations: 25
```

Logistic Regression for All Predictors

```
> #if we do it for each, only money that is significant
> logit_dem_money<- glm((win_dem) ~ money_raised_share_dem,
+                       data = data, family = "binomial")
> summary(logit_dem_money)

Call:
glm(formula = (win_dem) ~ money_raised_share_dem, family = "binomial",
    data = data)

Coefficients:
                       Estimate Std. Error z value Pr(>|z|)
(Intercept)             -20.161      7.126  -2.829  0.00467 **
money_raised_share_dem   34.378     12.104   2.840  0.00451 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 66.406  on 49  degrees of freedom
Residual deviance: 10.707  on 48  degrees of freedom
AIC: 14.707

Number of Fisher Scoring iterations: 8
```

Logistic regression for money

```
> exp(coef(logit_dem_money))
          (Intercept) money_raised_share_dem
         1.754298e-09           8.512261e+14
```

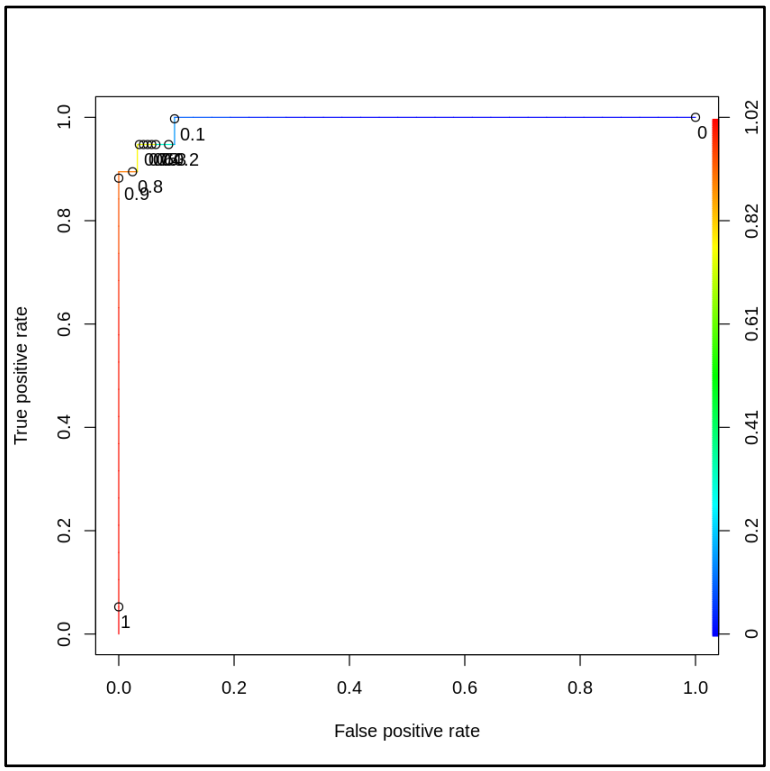
Odds ratios

```
> summary(predict)
     Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
0.0000028 0.0002491 0.0393527 0.3800000 0.9632595 0.9999728
```

Predicted values

```
> table(data$win_dem, predict > 0.5)

   FALSE TRUE
  0    30    1
  1     1   18
```

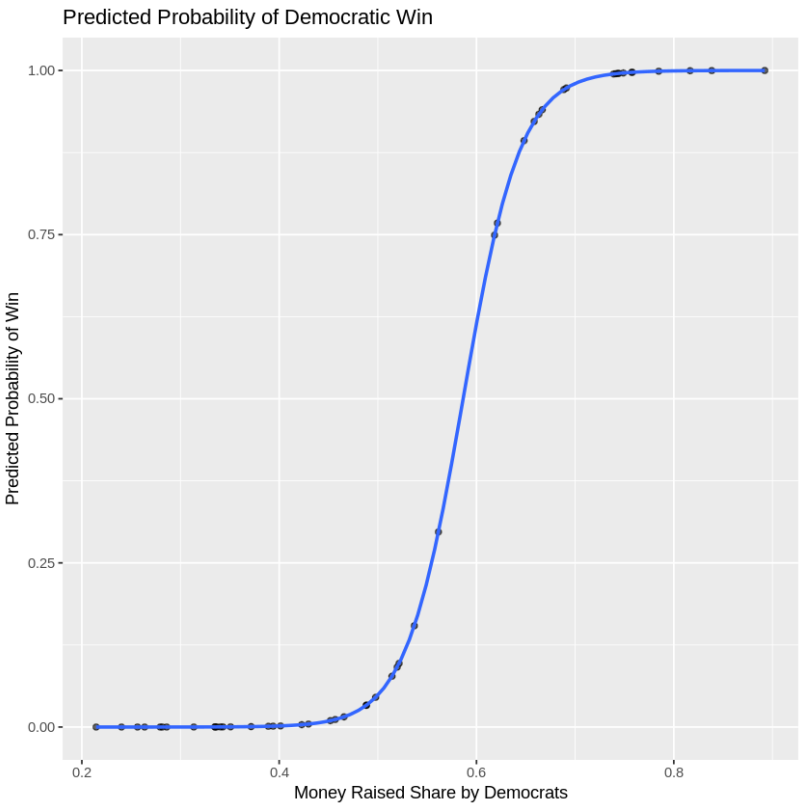
Predicted values



ROC Curve

```
> auc(data$win_dem,data$money_raised_share_dem)
Setting levels: control = 0, case = 1
Setting direction: controls < cases
Area under the curve: 0.9932
```

Area Under the Curve



Predicted Probability Graph

REPUBLICAN

```
> #overall model (which has terrible p value)
> logit_rep<- glm((win_rep) ~ money_raised_share_rep + unemployment_rate + crime_rate + education_rate + population,
+                 data = data, family = "binomial")
Warning messages:
1: glm.fit: algorithm did not converge
2: glm.fit: fitted probabilities numerically 0 or 1 occurred
> summary(logit_rep)

Call:
glm(formula = (win_rep) ~ money_raised_share_rep + unemployment_rate +
    crime_rate + education_rate + population, family = "binomial",
    data = data)

Coefficients:
                         Estimate Std. Error z value Pr(>|z|)
(Intercept)            -1.000e+03  3.098e+05  -0.003    0.997
money_raised_share_rep  1.805e+03  5.208e+05   0.003    0.997
unemployment_rate       7.479e+03  2.401e+06   0.003    0.998
crime_rate              1.682e+03  6.560e+05   0.003    0.998
education_rate         -3.396e+02  1.683e+05  -0.002    0.998
population              3.978e-06  1.397e-03   0.003    0.998

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 6.6406e+01  on 49  degrees of freedom
Residual deviance: 2.2811e-08  on 44  degrees of freedom
AIC: 12

Number of Fisher Scoring iterations: 25
```

Logistic Regression for All Predictors

```
> #if we do it for each, only money that is significant
> logit_rep_money<- glm((win_rep) ~ money_raised_share_rep,
+                       data = data, family = "binomial")
> summary(logit_rep_money)

Call:
glm(formula = (win_rep) ~ money_raised_share_rep, family = "binomial",
    data = data)

Coefficients:
                      Estimate Std. Error z value Pr(>|z|)
(Intercept)            -14.217      5.093  -2.792  0.00524 **
money_raised_share_rep  34.378     12.104   2.840  0.00451 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 66.406  on 49  degrees of freedom
Residual deviance: 10.707  on 48  degrees of freedom
AIC: 14.707

Number of Fisher Scoring iterations: 8
```

Logistic Regression for Money

```
> exp(coef(logit_rep_money))
           (Intercept) money_raised_share_rep
          6.696560e-07           8.512261e+14
```

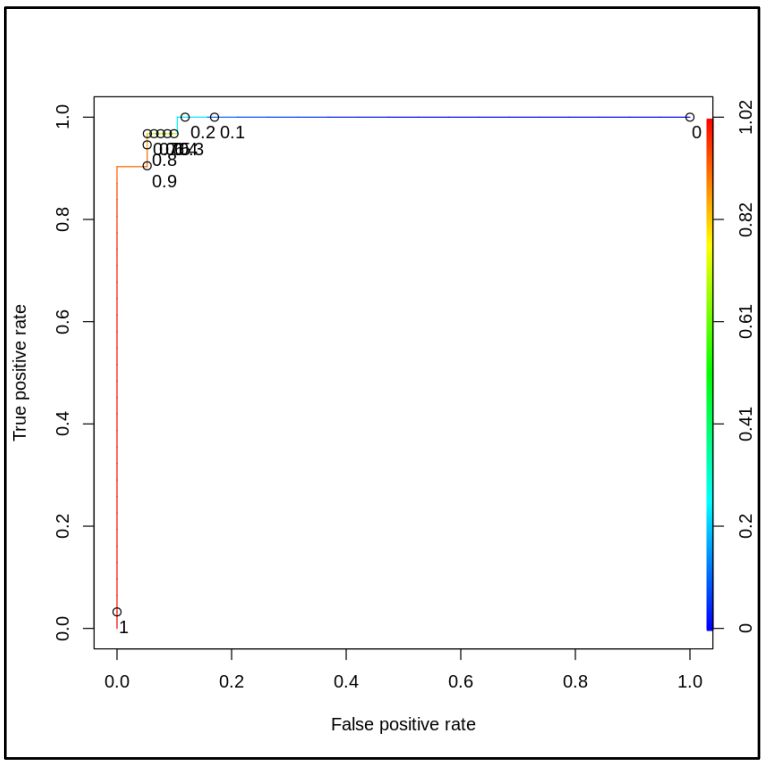
Odds ratio

```
> summary(predict)
     Min.   1st Qu.    Median      Mean   3rd Qu.      Max. 
0.0000272 0.0367405 0.9606473 0.6200000 0.9997509 0.9999972 
```

Predicted values

```
> table(data$win_rep, predict > 0.5)

    FALSE TRUE
  0    18    1
  1     1   30
```

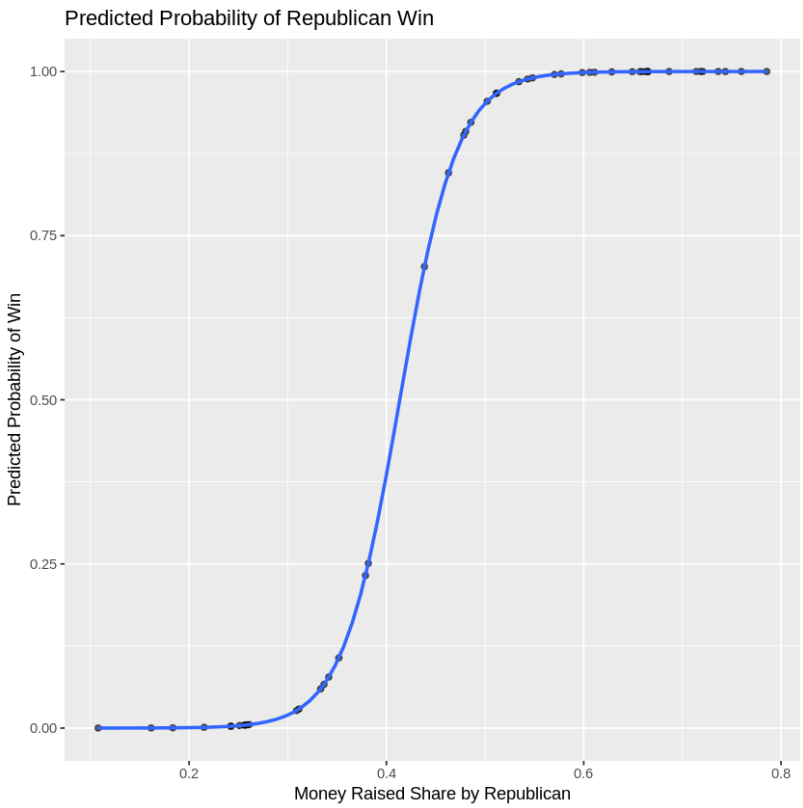
Confusion Matrix



ROC Curve

```
> auc(data$win_rep,data$money_raised_share_rep)
Setting levels: control = 0, case = 1
Setting direction: controls < cases
Area under the curve: 0.9932
```

Area Under the Curve



Predicted Probability Graph

VII. Discussion

Discussion for the Linear Regression and Heteroskedasticity (Democratic Party)

Based on the linear regression, the model has a good fit to the data since it has a high adjusted R squared value of 0.9685, meaning the model explains approximately 96.85% variance in the data. The overall F-test p-value (< 2.2e-16) < 0.05, indicating that the model is statistically significant overall. Each of the predictors also has a relatively low VIF value (< 4), possibly indicating low multicollinearity among the independent variables. However, not all of the predictors are statistically significant in predicting the vote result for the democratic party (the unemployment_rate p-value (0.3155) > 0.05). There is also indication of multicollinearity, shown by the flipped sign between correlation matrix and the estimated coefficient in the model for the variable crime_rate (-8.920e+06 (cor) vs 0.07504 (model)). We can also see from the correlation matrix that there are correlation values stronger than 0.6 between votes_dem and money_raised_dem (0.89913), between population and votes_dem (0.971290), between population and money_raised_dem (0.84840525), indicating strong multicollinearity and potential redundancy, may causing the problem of our analysis. Furthermore, as we can see from the residual vs fitted plot, there is curvature in the red line and the residuals are not evenly distributed across the fitted value (there were also some noticeable outliers), which indicates heteroskedasticity in our model. This supported by the result from the Breusch-Pagan test statistics (Ho: the residuals have constant variance (homoscedasticity), Ha: the residuals have non-constant variance (heteroskedasticity) that shows the p-value (5.049e-05) < 0.05, which means that we reject the null hypothesis and there is sufficient evidence to conclude that our model has heteroskedasticity.

Discussion for the Linear Regression (Republican Party)

Based on the linear regression analysis, the Republican Party model exhibits a strong fit to the data, as evidenced by a high adjusted R-squared value of 0.9169, indicating that approximately 91.69% of the variance in vote outcome for Republican party can be explained by the model variables. The overall F-test is highly significant (p-value < 2.2e-16), confirming that the model is statistically significant. Each predictor also has a VIF value below the threshold of 5, indicating low levels of multicollinearity amongst the independent variables. This is further supported when checking for the coefficient signs in the correlation matrix and correlation regression which all matched suitably. However, not all predictors are statistically significant in predicting voting results for the Republican party such as unemployment_rate with a p-value > 0.05 (0.9612). Moreover, there are some correlations above 0.6 value between variables such as money_raised_rep and population (0.848229), indicating strong multicollinearity and potential redundancy that may cause the generalization of our overall findings. Furthermore, the residual vs. fitted plot shows similar structure to the one we found on the Democrat, there is a curvature in the red line and residuals are not evenly dispersed across the fitted value with some noticeable outliers - overall indicating heteroskedasticity in our Republican model. Moreover, this is supported by the results from the Breusch-Pagan test statistic which shows a p-value of 1.31e-05 < 0.05, which indicates that we reject the null hypothesis of homoskedasticity and hence, possess enough evidence to conclude that our model indeed suffer from heteroskedasticity.

Discussion for the Logistic Regression (Democrats)

Since the logistic regression result was not as desired for all of the variables, we only analyzed the most significant predictor, which is the money raised (money_raised_share), and whether it is a significant predictor in winning for each party. For the Democratic, the odds ratio for the coefficient of money raised is extremely large, indicating that the higher money raised by Democrats is strongly associated with the democratic win. However, this extreme odds ratio happened due to high correlation with the outcome and lack of other important predictors in the model, which may make the model overfit the training data. The predicted probabilities range from 0.0000028 to 0.9999728, covering almost the full range of possible probabilities and suggest that money raised provides a clear separation between winning (win=1) and loss (win=0). Based on the confusion matrix, we get a specificity of 94.74% means that the model correctly identifies 94.74% of actual wins (true positives). For the sensitivity, we get a value of 96.77% which means that the model correctly identifies 96.77% of actual losses (true negatives). Therefore, the overall accuracy is 96%, which means that the model correctly classifies 96% of the data. Based on the ROC curve, we get the area under curve (AUC) of 0.993, indicating that the logistic model for democrats almost perfectly separates winners from losers based on the money raised. Moreover, the close alignment of most data points with the predicted probability curve suggests that the logistic regression model effectively captures the influence of fundraising on election outcomes, confirming that money raised is a significant predictor of success for the Democratic Party. However, the range of the money raised value appears slightly broader compared to the Republican, indicate a wider spread in fundraising share for Democrats.

Discussion for the Logistic Regression (Republicans)

Similar to the democrats, since the result for logistic regression was not desired for all variables, we will be focusing on analyzing the most significant predictor for the Republican party, which is the money raised (mone_raised_share) to determine its significance as a predictor in winning for this party. For the Republicans, the odds ratio for coefficient of money is also significantly high. This indicates that higher money raised by Republicans is strongly associated with a Republican win. Yet, this excessively high odds ratio results from high correlation with the election outcomes and lack of other significant predictors in the model, potentially leading to overfitting of the training data. The predicted probabilities from the logistic regression for the Republicans range from 0.0000272 to 0.9999972, indicating a broad coverage of possible outcomes and suggesting that money raised provides a distinct separation between winning (win = 1) and losing (win = 0). From the confusion matrix, the sensitivity of the model is 96.77%, meaning that it correctly identifies 96.77% of actual wins (true positive). The specificity is 94.74%, indicating that the model accurately identifies 94.74% of actual losses (true negatives). Consequently, the overall accuracy of the model is 96% which signifies that it classifies 96% of the observations correctly. From the ROC curve analysis, the area under the curve (AUC) for the Republican model is 0.9932, which suggests that the logistic model for Republicans nearly perfectly separates winners from losers based on the money raised, similar to the Democratic model but with slightly more sensitivity and less specificity. Furthermore, the predicted probability graph illustrates a sharp transition from low to high predicted probabilities of a Republican win as the money raised increases, clearly delineating between different levels of fundraising. This indicates a strong correlation between the amount of money raised and the likelihood of winning. Additionally, the close alignment of most data

points with the predicted probability curve suggests that the logistic regression model effectively captures the influence of fundraising on election outcomes, confirming that money raised is a significant predictor of success for the Republican Party.

VIII. Conclusion

This investigation sought to identify the most influential predictors of U.S. presidential election outcomes by analyzing socio-economic and demographic variables via statistical modelling in RStudio. Our analysis revealed that money raised by each political party is the most significant and reliable predictor of electoral success, bearing strong correlations with vote counts for both democrats and republicans. Conducting a logistic regression, we further confirmed the predictive power of this variable which yielded 96% accuracy rates in correctly classifying election winners and losers. The other variables we explored – unemployment rate, crime rate, education rate, and population – exhibited mixed results. While education rate did demonstrate a moderately positive correlation with democratic votes, as we expected, its value was inconsistent across our models as it did not bear significance in republican votes. Similarly, population was positively correlated with votes for both parties, but suffered from multicollinearity with money raised, as mentioned in section V11. The variables of unemployment and crime rates showed limited predictive power, with unemployment rate consistently presenting as statistically insignificant in our models and crime rate having inconsistent effects across models, indicating that these variables may play a secondary role in shaping electoral outcomes.

While still fruitful, our study faced limitations, including multicollinearity among predictors and potential biases in collected data. Future research could benefit from incorporating additional variables into the study (such as income inequality or regional

cultural factors) to develop a more nuanced understanding of the influencing factors in voter behaviour.

To conclude our study, our findings point to the outsized influence of financial resources on U.S. presidential elections, reinforcing the central role of campaign fundraising in modern political battles.

IX. References

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